

Software Engineering: Requirements Engineering & UML Modelling
Problem Statement #27 | Campus & Academic Operations: Smart Cities, Transport & Logistics Warehouse Inventory & Pallet Location Tracker

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**PROBLEM STATEMENT #27 | Smart Cities,
Transport & Logistics Warehouse Inventory
& Pallet Location Tracker**

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1.Functional Requirements (FR-001 to FR-005)

Req ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall validate proposed pallet placement coordinates against the rack's maximum load capacity before confirming storage.	High	Pass: Pallet is assigned to a valid rack where the total weight remains within the rated structural limit. Fail: System prevents placement when the total weight would exceed the rack's maximum capacity.	Prevents rack overloading and ensures safe and structurally compliant pallet storage.
FR-002	The system shall allow the Warehouse Operator to record and maintain the 3D location of each pallet using aisle, rack, level, and slot coordinates.	High	Pass: Pallet is stored with valid aisle, rack, level, and slot coordinates. Fail: System rejects storage when the location is invalid, unavailable, or incomplete.	Ensures accurate pallet positioning and enables efficient inventory retrieval.
FR-003	The system shall identify pallets and record stock movements using barcode or RFID scanning for inbound, outbound, and transfer operations.	High	Pass: Valid barcode/RFID scan identifies the pallet and records the corresponding stock movement. Fail: System rejects invalid or duplicate scans and displays an appropriate error message.	Automates inventory data capture, reduces manual entry errors, and improves movement accuracy.
FR-004	The system shall maintain a time-stamped history of stock movements including pallet ID, movement type, Warehouse Operator ID, and source and destination locations.	Medium	Pass: Each completed movement is stored with all required details and a timestamp. Fail: System does not complete the movement record when critical information is missing.	Provides traceability and an audit trail for all pallet movements within the warehouse.
FR-005	The system shall allow authorized users to search for an SKU or pallet ID and display its current 3D warehouse location and inventory status.	Medium	Pass: A valid SKU or pallet ID returns the correct current location and inventory status. Fail: System displays an appropriate message when no matching pallet is found.	Enables warehouse personnel to quickly locate inventory and improves operational efficiency.

2. Non-Functional Requirements (NFR-001 & NFR-002)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Scalability	The warehouse item lookup service shall locate and return the current SKU pallet coordinate across 100,000 bins within 100 ms under simulated peak-load conditions.	High	Pass: Benchmarking confirms that SKU/pallet location queries return results within 100 ms across 100,000 bins under peak-load testing. Fail: Query response time exceeds 100 ms during the required benchmark.	Ensures warehouse operators and supervisors can locate inventory quickly even when the system manages a large number of storage bins.
NFR-002	Security & Access Control	The system shall enforce Role-Based Access Control (RBAC) for Warehouse Operators and Logistics Supervisors and encrypt sensitive inventory and user data both at rest and in transit.	High	Pass: Users can access only functions permitted by their assigned role, and encrypted communication is verified during security testing. Fail: An unauthorized user can access restricted inventory functions or sensitive data.	Protects inventory information, prevents unauthorized modifications, and ensures secure access to warehouse operations.